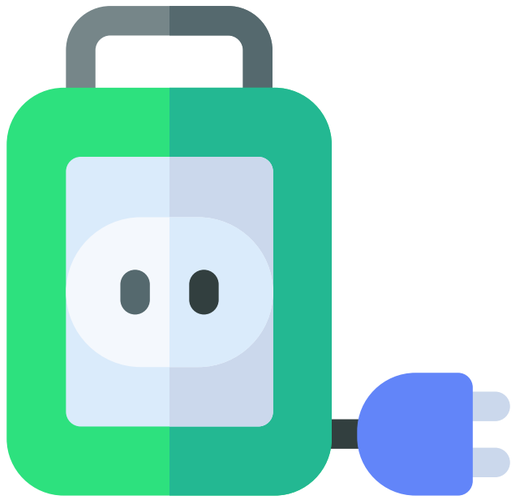
Introduction

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SMART Wall Socket

ESP32 / Firebase / Mobile App

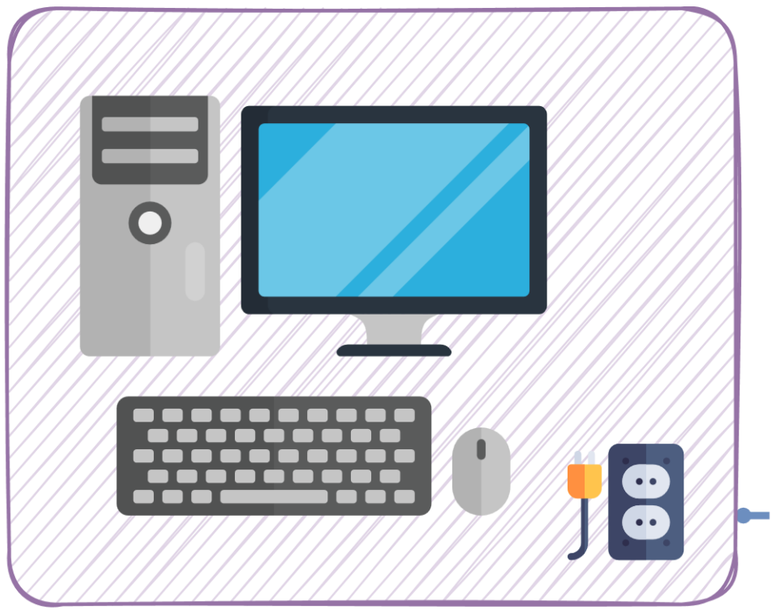
* Remote Control Socket (On/Off) Through Android App
* Electricity Usage Monitoring
* Bill Calculation Based on Usage
* Analyze Usage Patterns

Internet of Things (IoT) Project Development

System Design

SMART Wall Socket

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Power Consumption Device / Workdesk



Smart Wall Plug

* On/Off Plug Wirelessly
* ESP32 Powered
* Usage Record History



Wall Socket



SMART Wall Socket



Step 1 → Scan QR



Step 2 → Pair & Control



Step 3 → Monitor Usage



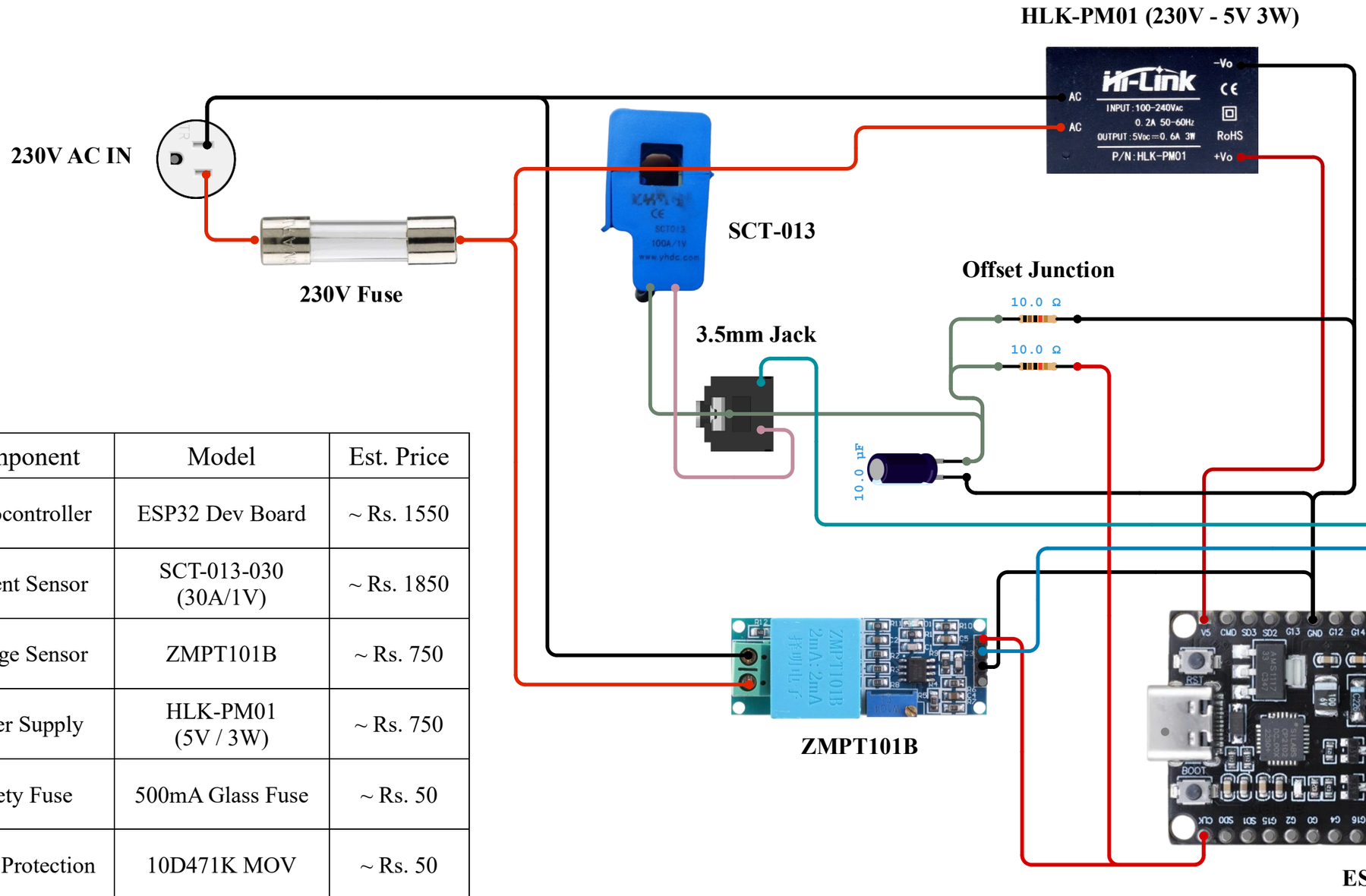
* On/Off Socket Remotely
* Monitor Electricity Usage
* Calculate Price in Rupees
* Analyze Usage Patterns

Internet of Things (IoT) Project Development

Circuit Diagram

SMART Wall Socket

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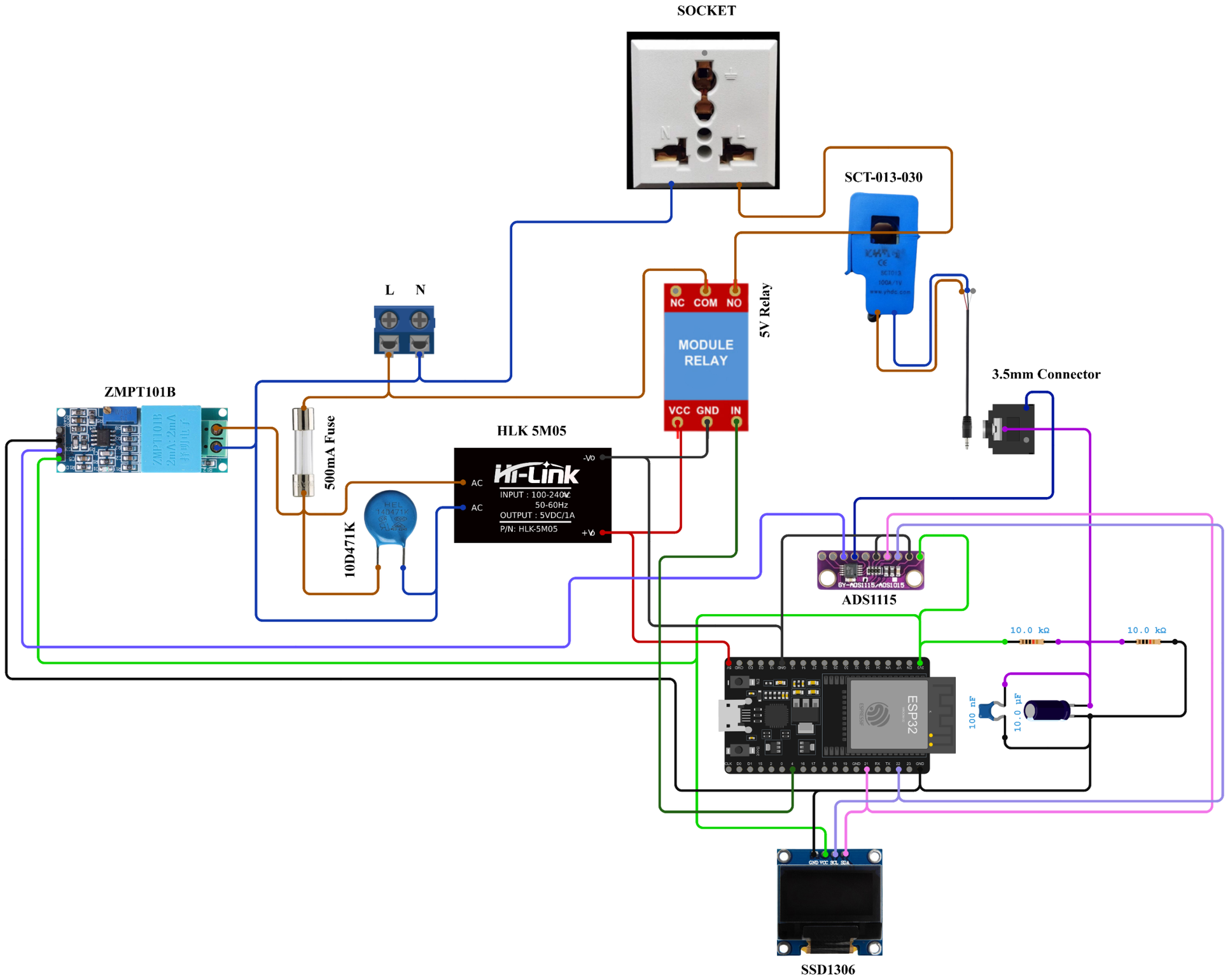
Component	Model	Est. Price
Microcontroller	ESP32 Dev Board	~ Rs. 1550
Current Sensor	SCT-013-030 (30A/1V)	~ Rs. 1850
Voltage Sensor	ZMPT101B	~ Rs. 750
Power Supply	HLK-PM01 (5V / 3W)	~ Rs. 750
Safety Fuse	500mA Glass Fuse	~ Rs. 50
Surge Protection	10D471K MOV	~ Rs. 50
Offset Junction	2× 10kΩ Resistors 1× 10µF Capacitor	~ Rs. 50
Connector	3.5mm Female Audio Jack	~ Rs. 150

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Circuit Diagram - Master

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Internet of Things (IoT) Project Development

Circuit Diagram - Slave

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